

Siddhant Mohan

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EDUCATION

- **New York University Tandon (NYU) School of Engineering** New York City, USA
• *Master of Science (MS) - Electrical Engineering; CGPA: 3.7/4* Aug. 2024 - May 2026
Courses: Efficient AI, Advanced Computer Vision, Natural Language Processing, ML Systems Engineering & Operations
- **Indian Institute of Technology (IIT) Tirupati** Tirupati, India
• *Bachelor of Technology (B. Tech) - Electrical Engineering; CGPA: 8.0/10* Jul. 2016 - Jun. 2020
Courses: Deep Learning, Industrial Data Science, Medical Imaging, Digital Signal Processing

TECHNICAL SKILLS

- **Areas of Interest:** Artificial Intelligence, Machine Learning, Computer Vision, Natural Language Processing (NLP)
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, HuggingFace, OpenCV, OpenMMLab, MLFlow, Ray, Scikit-Learn, ONNX, Numpy, Scipy, Pandas, Matplotlib
- **MLOps & Cloud:** Git, Docker, Singularity, Slurm, Kubernetes, ArgoCD, Prometheus, MinIO, Terraform, Grafana, GCP
- **Model Architectures:** Transformers, Large Language Models (LLMs), Convolutional Neural Networks, Diffusion Models
- **Techniques:** Unsupervised Learning, Fine-Tuning, Data & Model Parallelism, Statistics, Probability, Data Engineering
- **Programming Languages:** Python, C++, C, MATLAB, SQL

WORK EXPERIENCE

- **Toshiba Software India - R&D Division** Bengaluru, India
• *Senior Research Engineer - Computer Vision Team* Jul. 2023 - Jul. 2024
Research Engineer Jul. 2020 - Jul. 2023
 - **Incremental Learning:** Engineered an **incremental learning** pipeline for a single-shot object detector model, using **knowledge distillation** and **weight importance-based regularization**, which enabled adaptation to new object classes without forgetting old classes, improving mAP by 15% on COCO and VOC.
 - **Unsupervised Anomaly Localization:** Directed a team in developing an unsupervised approach for localizing anomalies in industrial images using **feature extractors** and **k-nearest neighbor statistics**. Outperformed baselines on the **MVTec-AD** dataset by 22%.
 - **Domain Adaptation:** Managed a project on implementing domain adaptation for **key-point detection** in arc-welding images, using **mix-up augmentation**. Achieved an improvement of 12% over the baseline.
 - **Systems & Infrastructure:** Configured and deployed NVIDIA GPU workstations from scratch to set up an **R&D lab**, including **Ubuntu**, **CUDA**, **cuDNN driver setup**, **PyTorch environments** for deep learning, reducing compute costs by 30% annually compared to equivalent commercial cloud usage.
- **UnboundX** Short Hills, NJ
• *AI Engineering Intern* Jun. 2025 - Aug. 2025
 - Led a team to develop **feed-ranking recommender systems** for an investment-focused **social media** app, combining a time-decay based scoring model with BERT-based content classification to generate personalized, category-driven feeds. Utilized vectorized operations to reduce ranking latency by 10%.
- **Tech Mahindra (formerly ZEN3 Infosolutions)** Hyderabad, India
• *Machine Learning Intern* May 2019 - Jul. 2019
 - Devised, trained and deployed a customer support **chatbot** for mobile data plans using **Recurrent Neural Networks**, within 8 weeks. Facilitated the creation of a custom Q&A dataset for training.

ACADEMIC RESEARCH PROJECTS

- **Efficient Data Pipelines for Vision-Language Models:** As part of SAI lab under Prof. Sai Qian Zhang, improved computational efficiency of a **Hybrid Autoregressive Transformer (HART)** using VQ-VAEs for vision-language generation by 20% by performing semantic clustering of text prompts and reusing low-resolution intermediate image embeddings within clusters. Reduced forward passes, optimized multi-GPU inference, and lowered compute and memory overhead in large-scale visual generation pipelines. (Sep 2025).
- **Distribution-Aware Companding Quantization (DACQ):** Designed and implemented a **LLM post-training quantization framework** that models layer-wise weight distributions and applies non-uniform companding to minimize accuracy degradation under aggressive INT4 settings to enable low-bit inference. Built multi-GPU evaluation pipelines to benchmark perplexity, memory footprint, and throughput on LLaMa-3-8B and Qwen2.5-7B. ArXiv paper (Sep 2025).
- **Diffusion-Based Image Compression (DBIC):** Designed a two-stage **learned image compression** framework combining statistical encoding with **guided diffusion-based reconstruction** for perceptual quality enhancement. Transmitted compact structural representations and restored high-frequency detail using generative priors, optimizing the trade-off between bitrate, semantic fidelity, and perceptual image quality. Evaluated using reconstruction and perceptual metrics. (Sep 2025).
- **Intelligent Multimedia Processing Module (IMP):** Built an end-to-end pipeline to extract text and metadata from multimodal streams (audio, video, documents) and integrated a **Retrieval Augmented Generation (RAG) system with LLMs** for enterprise query resolution. Automated data extraction, semantic indexing, model deployment, and CI/CD workflows for scalable document search. Implemented vector storage and retrieval using **FAISS** for efficient similarity search across large-scale enterprise knowledge bases. (May 2025).